

Operating Instructions

Diesel engine
12 V 4000 M90
16 V 4000 M90

MW15413/06E



Power. Passion. Partnership.

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Commissioning Note

Important

Please complete and return the "Commissioning Note" card below to MTU Friedrichshafen GmbH.
The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).



Postcard

MTU Friedrichshafen GmbH
Technical Information Management
Dept. VOT
88040 Friedrichshafen
GERMANY

Engine No.:

MTU works order no.:

Engine model:

Date put into operation:

Application: *	
☐ Marine	☐ Rail
☐ Genset	☐

Type:
Manufacturer:

End user's address:

Remarks:

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1 Safety

1.1 General conditions

General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention and environmental protection must be observed. This state-of-the-art engine has been designed to meet all applicable laws and regulations. The engine may nevertheless present a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the Safety Instructions

Correct use

The engine is intended solely for use in accordance with contractual agreements and the purpose envisaged for it on delivery. Any other use is considered improper use. The engine manufacturer accepts no liability whatsoever for resultant damage or injury in such case. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the operating instructions and maintenance and repair specifications.

Modifications or conversions

Unauthorized modifications to the engine represent a safety risk.

MTU will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

Spare parts

Only genuine MTU spare parts must be used to replace components or assemblies. MTU accepts no liability whatsoever for damage or injury resulting from the use of other spare parts and the warranty shall be voided in such case.

Reworking components

Repair or engine overhaul must be carried out in workshops authorized by MTU.

1.2 Personnel and organizational requirements

Personnel requirements

Work on the engine must only be carried out by appropriately qualified and instructed personnel.

Observe the minimum legal age.

Responsibilities of the operating, maintenance and repair personnel must be specified by the operating company.

Organizational measures

This publication must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep it handy in the vicinity of the engine such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use the manual as a basis for instructing personnel on engine operation and repair. In particular, personnel must have read and understood the safety-relevant instructions.

This is especially important for personnel who work on the engine only on an occasional basis. These persons shall receive repeated instruction.

Use the Spare Parts Catalog to identify spare parts during maintenance and repair work.

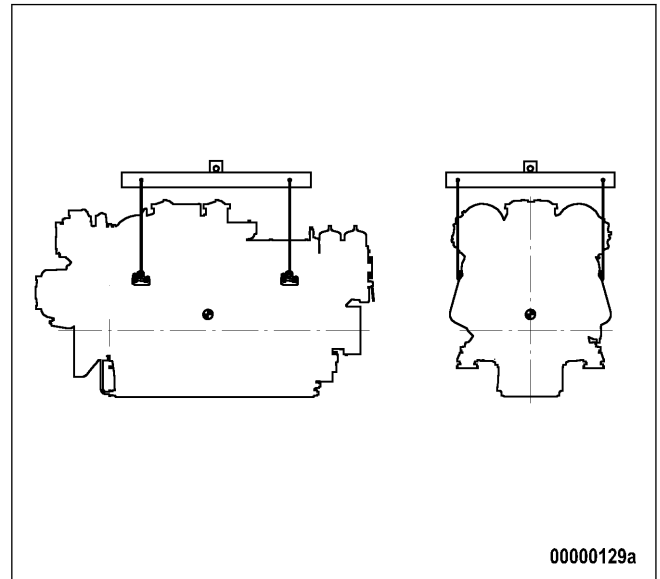
Working clothes and protective equipment

Wear proper protective clothing for all work.

Depending on the kind of work, use the necessary personal protective equipment.

1.3 Transport

Transport



Lift the engine only with the lifting eyes provided.

The lifting eyes are designed for engine transport only, not for the transport of propulsion units (engine and gearbox).

Use only the transport and lifting equipment approved by MTU.

The engine must only be transported in installation position, max. permissible diagonal pull 10°.

Take note of the engine center of gravity.

In the case of special packaging with aluminum foil, suspend the engine on the lifting eyes of the transport pallet or transport with equipment for heavy loads (forklift truck).

Prior to transporting the engine, it is imperative to install transportation locking devices for crankshaft and engine mounts.

Secure the engine against tilting during transportation. The engine must be especially secured against slipping or tilting when going up or down inclines and ramps.

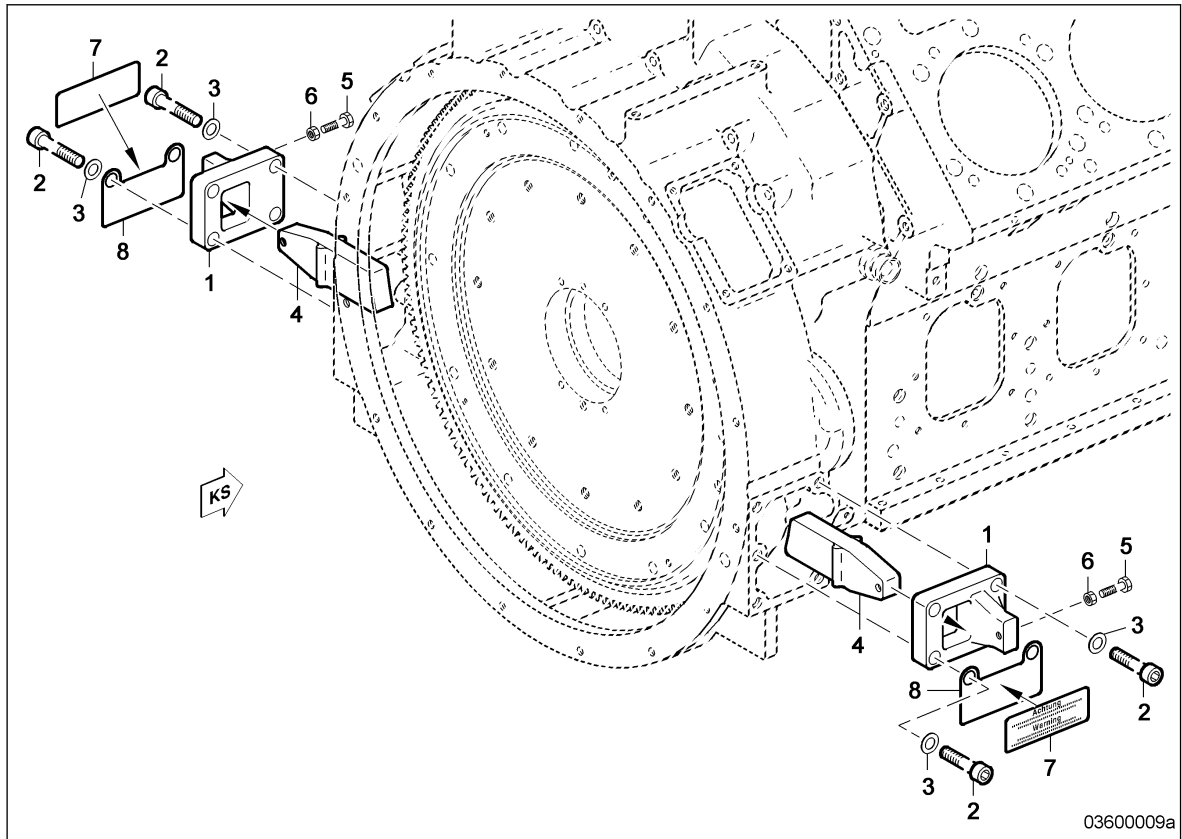
Setting the engine down after transport

Place the engine only on an even, firm surface.

Ensure appropriate consistency and load-bearing capacity of the ground or support surface.

Never place an engine on the oil pan, unless expressly authorized by MTU on a case-to-case basis to do so.

1.4 Crankshaft transport locking device



Transport locking device

This locking device protects the crankshaft bearings from shocks and vibration damage during engine transport.

For removal of the transport locking device, observe the following:

- The transport locking device should remain installed as long as possible during engine installation in order to avoid damage.
- Starting or barring the engine is allowed only with the transport locking device removed. If the generator is already mounted on the engine, ensure that the transport locking device of the generator is removed.
- Prior to every engine transport (without generator), this transport locking device must be reinstalled according to the instructions.

Removing the transport locking device from driving end (KS)

- Release the locknuts (6) on both sides of the flywheel housing, remove screws (5) and take off the two locks (4).
- Remove screws (2), washers (3), cover plate (8) with label (7) and plates (1).

Store the removed parts of the transport locking device carefully for possible reuse!